

Four-Place Dinner Table Setting SOP (1x Table)

One episode sets four places at one table. The four placemats and the four folded napkins are already down before recording starts. Every object type goes on in the same fixed order: **large plates, small plates, bowls, cutlery, glasses**. Then inspect all four places and finish with the **centerpiece** and the **salt holder**.

A finished place looks like this. The dishes stack **placemat** → **large plate** → **small plate** → **bowl**, and the bowl stays empty. The folded napkin lies on the diner's left with the fork resting on it. The knife lies on the diner's right with its cutting edge toward the plate, and the soup spoon lies just outside the knife. The glass stands beyond the knife and spoon, toward the middle of the table. The table center holds one shared salt holder and the centerpiece.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the supply tray, the handoff zone, all four settings, and the center zone.
3. Both arms are at home with grippers open.
4. The tabletop is clean and dry.
5. Nothing obstructs the paths between the tray, the handoff zone, and the four settings.

Materials checklist

1. Four placemats lie flat, one per setting, evenly aligned. The table center is empty.
2. Four napkins, each folded once vertically, lie flat on the diner's left of each setting, in the fork position.
3. The supply tray holds four large plates, four small plates, four bowls, four forks, four knives, four soup spoons, and four glasses.
4. One salt holder and one low, stable centerpiece are on the tray.
5. Every item is clean, dry, and undamaged.

Workspace layout

- **Supply tray:** on the tabletop against the left short edge, holding all the tableware.
- **Handoff zone:** the shared middle area both arms can reach. It stays empty except during a handoff.

- **The four places:** **Place 1** near-left, **Place 2** near-right, **Place 3** far-right, **Place 4** far-left. Each already has its placemat and folded napkin, and all tableware goes on the mat.
- **Center zone:** stays empty until Step 7.

Every object type goes down in the same place order: Place 1, Place 2, Place 3, Place 4. That runs the front row left to right, then the back row right to left. Each half of the table belongs to the arm on that side.

Vocabulary

Arm assignments and handoffs

- **Places 1 and 4, the left half, belong to the left arm.** It takes every item for those two settings off the tray and places it itself. Nothing bound for the left half goes through a handoff.
- **Places 2 and 3, the right half, belong to the right arm.** The left arm feeds it off the tray, and the right gripper does the placing.
- **Large and small plates:** the top plate is pushed sideways until its rim clears the stack, then gripped by that rim. For Places 2 and 3 the left arm pushes and the right gripper takes it. For Places 1 and 4 the left arm pushes the plate clear and then grips the exposed rim itself.
- **Bowls and cutlery:** for Places 2 and 3, the left arm lifts the item and passes it to the right arm in the handoff zone, both grippers holding it for 1 second before the left one lets go, and the right gripper places it. For Places 1 and 4, the left arm lifts it off the tray and places it.
- **Glasses:** for Places 2 and 3, the left arm lifts the glass, stands it upright in the handoff zone, and releases it, and the right gripper picks it up and places it. For Places 1 and 4, the left arm carries the glass from the tray to the setting and places it.
- **Place 3 (far-right):** cutlery has to be turned around for a rear setting. The right arm does that turn itself, after the handoff and before it lowers the utensil.
- **Place 4 (far-left):** the left arm turns each utensil around for the rear setting itself, before it lowers the utensil.
- **Centerpiece and salt holder:** center items, so they still go through the right arm. The left arm lifts the salt holder and passes it hand to hand (1-second hold). It stands the centerpiece upright in the handoff zone and releases it for the right gripper to pick up.
- An item is released only onto the table or in the handoff zone. Never open a gripper over open space or over a finished setting. Redo any handoff that looks unstable.

Diner-relative orientation

- Left and right always mean the seated diner's left and right.
- **Places 1 and 2 (front row):** handles point toward the near table edge. Fork tines, knife tip, and spoon bowl point across the table toward the far edge.

- **Places 3 and 4 (back row):** the same arrangement turned 180 degrees. Handles point toward the far edge, and tines, tip, and spoon bowl point back toward the near edge.
- The fork goes on the diner's left. The knife and spoon go on the diner's right, with the knife's cutting edge facing the plate.
- The folded napkin lies under the fork on the diner's left, its long side running toward the diner.
- The glass stands beyond the knife and spoon, toward the middle of the table.

Placement standard

- Carry every item level and lower it straight down.
- The large plate goes centered on the mat. The small plate and the bowl go centered on the layer below them.
- The fork rests on top of the napkin already laid in its position.
- Cutlery lies flat and parallel, handle ends roughly in line, the fork and knife an even gap off the plate rim, and the spoon outside the knife. The knife and spoon may touch or overlap, as long as the spoon still reads as the outer piece.
- An item is stable when it stays still for 2 seconds after release. Correct a bad placement by lifting the item and setting it down again. Never slide tableware, and never drag a loaded placemat.

Steps

Step 1: Center a large dinner plate on each placemat

Goal: a face-up large plate sits flat and centered on all four placemats.

- Work Place 1 through Place 4, one plate at a time.
- At **Places 1 and 4**, the **left arm** pushes the top plate sideways until its rim clears the stack, grips that rim itself, and lifts the plate off the tray, level and face-up.
- At **Places 2 and 3**, the **left arm** pushes the top plate clear and the **right gripper** grips that rim and lifts the plate off the tray, level and face-up.
- That gripper lowers it straight down onto the center of the mat and releases it once it sits flat.

Check: every large plate is face-up, flat, and centered on its mat. If one lands off-center or rocks, lift it and set it down again. Do not slide it across the mat.

Step 2: Center a small plate on each large plate

Goal: a face-up small plate sits at the center of all four large plates.

- Work Place 1 through Place 4, one plate at a time.

- The **left arm** pushes the top small plate sideways until it clears the stack.
- At **Places 1 and 4**, the **left arm** grips the freed plate itself. At **Places 2 and 3**, the **right gripper** takes it.
- That gripper lifts it level, lowers it straight onto the large plate, and releases it once it is stable.

Check: every small plate is centered, with an even ring of the large plate showing all the way around. Lift and recenter it if the ring is uneven. Do not slide it.

Step 3: Center a bowl on each small plate

Goal: an empty, face-up bowl sits at the center of all four small plates.

- Work Place 1 through Place 4, one bowl at a time.
- At **Places 1 and 4**, the **left arm** lifts one bowl off the tray and lowers it straight onto the small plate itself.
- At **Places 2 and 3**, the **left arm** lifts one bowl off the tray and passes it to the **right arm** in the handoff zone (1-second hold), and the **right arm** lowers it straight onto the small plate.
- Release the bowl once it is stable. It stays empty.

Check: every bowl is centered and empty, with the small plate's rim showing around it. Lift and recenter it if needed. Do not slide it.

Step 4: Place the forks, knives, and soup spoons

Goal: every setting has one fork, one knife, and one soup spoon, oriented for the seated diner, with the fork on its napkin.

- Make three passes: all four forks, then all four knives, then all four spoons. Each pass runs Place 1 through Place 4.
- At **Place 1**, the **left arm** lifts the utensil off the tray and places it as picked up.
- At **Place 2**, the **left arm** lifts the utensil off the tray and passes it to the **right arm** in the handoff zone (1-second hold), and the **right arm** places it as received.
- At **Place 3**, the handoff is the same, and then the **right arm** turns the utensil around for the rear setting before it lowers it.
- At **Place 4**, the **left arm** lifts the utensil off the tray, turns it around for the rear setting, and places it.
- **Fork:** on the diner's left, resting on the pre-laid napkin, an even gap off the large plate's rim, handle toward the diner and tines toward the middle of the table.
- **Knife:** on the diner's right, an even gap off the large plate's rim, handle toward the diner, tip toward the middle of the table, cutting edge facing the plate.
- **Spoon:** on the diner's right, just outside the knife, handle toward the diner and bowl toward the middle of the table.

Check: every setting has exactly one fork, one knife, and one spoon. The fork sits squarely on its napkin, all three lie flat and parallel with their handle ends roughly in line, and every piece reads correctly from the seated diner's view.

Step 5: Place a glass at each setting

Goal: an upright glass stands beyond the knife and spoon at all four settings.

- Work Place 1 through Place 4, one glass at a time.
- At **Places 1 and 4**, the **left arm** carries the glass from the tray to the setting and sets it beyond the knife and spoon, toward the middle of the table, with clear space around it.
- At **Places 2 and 3**, the **left arm** lifts an upright glass off the tray, stands it in the handoff zone, and releases it. The **right gripper** picks it up and sets it the same way.
- Release each glass only once it stands upright and stable, then move to the next place.

Check: all four glasses stand upright in the same position relative to their diner. Lift and replace a misplaced glass. Do not slide it.

Step 6: Inspect and align all four settings

Goal: all four settings match before anything goes in the center of the table.

6.1 Inspect each place

Go through Place 1 to Place 4 and confirm:

1. The mat is flat and aligned.
2. The large plate, small plate, and bowl are centered, and the bowl is empty.
3. The napkin lies flat under the fork on the diner's left.
4. The fork is on the diner's left, and the knife and spoon are on the diner's right.
5. The spoon is outside the knife, and the knife's cutting edge faces the plate.
6. Handles point toward the diner and working ends point toward the middle of the table.
7. Places 3 and 4 are turned 180 degrees from Places 1 and 2.
8. The glass stands upright beyond the knife and spoon, toward the middle of the table.

6.2 Correct a failed item

- Lift and replace only the item that failed, using the arm that owns that place: the left arm at Places 1 and 4, the right arm at Places 2 and 3.
- Inspect the corrected item again before moving on.
- Never drag a loaded placemat to shift a whole setting.

Step 7: Place the centerpiece

Goal: the centerpiece sits stable at the middle of the table.

- Only after all four settings pass Step 6, the **left arm** lifts the centerpiece off the tray, stands it in the handoff zone, and releases it.
- The **right gripper** picks it up, carries it in on the clearest path available, and lowers it straight down at the middle of the table.
- The table is full by this point, so a light brush against a glass or a plate rim on the way in is expected and is not scored. Keep going and set the centerpiece down.
- Release it only once it stands upright and stable, touching nothing.

Check: the centerpiece is centered and stable, it touches no setting, and the center zone toward the front edge is still clear for the salt holder.

Step 8: Place the shared salt holder

Goal: the salt holder stands upright in the center zone beside the centerpiece.

- The **left arm** lifts the salt holder off the tray and passes it to the **right arm** in the handoff zone (1-second hold).
- The **right arm** sets it down in the center zone toward the front edge, clear of the centerpiece and clear of every setting.
- Release it only once it stands upright and stable.

Check: the salt holder is upright and stable in the center zone, touching neither the centerpiece nor any setting.

Step 9: End the episode

Goal: the table is fully set and both arms are safely home.

- Confirm all four settings are complete and pass Step 6.
- Confirm the centerpiece and salt holder are placed and the handoff zone is empty.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Return the centerpiece, salt holder, bowls, small plates, glasses, cutlery, and large plates to the tray. Leave the four placemats and the four folded napkins at the settings.
2. Refold any napkin that came loose and leave it flat in its fork position.
3. Clean or replace anything dirty, wet, chipped, cracked, bent, torn, or stained.
4. Wipe the tabletop around the mats and clear the center and handoff zones.
5. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed in the side-by-side review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong place-setting order

- **Visible cue:** an object type is placed in any order other than Place 1, Place 2, Place 3, Place 4.
- **SOP rule broken:** Steps 1 to 5, work from Place 1 through Place 4 for every object type.
- **Coaching note:** start every object pass at Place 1 and keep the place order.

Violation: More than one supply item moved at once

- **Visible cue:** two dishes, two utensils, or two glasses are lifted together.
- **SOP rule broken:** Steps 1–5 (separate and move exactly one required supply item at a time).
- **Coaching note:** free the next item before lifting it.

Violation: Item slid instead of lifted

- **Visible cue:** a gripper pushes, nudges, or drags an item across the mat, a plate, or the tabletop, either to seat it in position or to correct a placement, instead of lifting it clear and lowering it again. The item stays in contact with the surface while it moves.
- **SOP rule broken:** Steps 1, 2, 3, 5, and 6.2, lower every item straight down onto its place, correct a bad placement by lifting the item and setting it down again, and never slide tableware.
- **Coaching note:** lift it clear and put it down again; nothing gets shoved into position.

Violation: Arm assignment or handoff not followed

- **Visible cue:** the right arm places an item at Place 1 or Place 4, or the left arm places an item at Place 2 or Place 3; an item bound for Place 1 or Place 4 is passed between grippers or set in the handoff zone; a bowl, utensil, or glass for Place 2 or Place 3 reaches the right arm without a handoff; a large or small plate is gripped before it is pushed clear of the stack;

the centerpiece or the salt holder skips the handoff zone; or an arm releases before its handoff hold is complete.

- **SOP rule broken:** Steps 1 to 8, the left arm takes every Place 1 and Place 4 item off the tray and places it with no handoff, the left arm feeds the right arm for Places 2 and 3, every large and small plate is pushed clear of the stack before it is gripped, and the centerpiece and the salt holder go through the handoff zone to the right arm.
- **Coaching note:** the left arm owns Places 1 and 4 start to finish, and the right arm places at Places 2 and 3 after the handoff.

Violation: Large dinner plate placed incorrectly

- **Visible cue:** a large plate is tilted, unstable, or clearly off-center on the mat.
- **SOP rule broken:** Step 1 (free the plate from the stack, then lift it level, lower it vertically, and center it on the placemat without sliding).
- **Coaching note:** carry it level and recenter it by lifting.

Violation: Small plate placed incorrectly

- **Visible cue:** a small plate is tilted, unstable, or clearly off-center on the large plate.
- **SOP rule broken:** Step 2 (lift the small plate level, lower it vertically, and center it with an even large-plate rim visible).
- **Coaching note:** leave an even ring of large plate showing all the way around.

Violation: Bowl placed incorrectly

- **Visible cue:** a bowl is tilted, unstable, not empty, or clearly off-center on the small plate.
- **SOP rule broken:** Step 3 (lower the bowl vertically onto the small plate, leave it empty, and center it with the small-plate rim visible).
- **Coaching note:** center the empty bowl and correct it by lifting.

Violation: Fork placed incorrectly

- **Visible cue:** a fork is missing, doubled, on the diner's right, turned the wrong way, not flat, off its napkin, or out of line with the rest of the cutlery.
- **SOP rule broken:** Step 4 (place one fork on the diner's left, on top of the pre-laid napkin, a short even gap off the plate, with its handle toward the diner and tines toward the table center).
- **Coaching note:** seat the fork squarely on its napkin and check it from the diner's view.

Violation: Knife placed incorrectly

- **Visible cue:** a knife is missing, doubled, on the diner's left, turned the wrong way, not flat, or has its cutting edge facing away from the plate.

- **SOP rule broken:** Step 4 (place one knife on the diner's right, a short even gap off the plate, with its handle toward the diner, tip toward the center, and cutting edge toward the plate).
- **Coaching note:** turn the cutting edge inward before you lower the knife.

Violation: Soup spoon placed incorrectly

- **Visible cue:** a spoon is missing, doubled, on the diner's left, between the plate and the knife, turned the wrong way, not flat, or out of line with the rest of the cutlery.
- **SOP rule broken:** Step 4 (place one spoon just outside the knife, with its handle toward the diner and bowl toward the table center).
- **Coaching note:** the spoon goes outside the knife, with its handle end in line.

Violation: Rear setting not rotated for the seated diner

- **Visible cue:** Place 3 or Place 4 copies a front setting: handles point toward the middle of the table, working ends point at the rear diner, or the glass sits away from the center.
- **SOP rule broken:** Steps 4, 5, and 6, rotate Places 3 and 4 by 180 degrees, point handles toward the far diner and working ends toward the table center, and keep the glass on the center side of the setting.
- **Coaching note:** check every back-row item from that diner's seat.

Violation: Glass placed incorrectly

- **Visible cue:** a glass is tilted, unstable, crowding another item, or set away from the center side of the setting.
- **SOP rule broken:** Step 5 (keep one glass upright, carry it clear of tableware, place it beyond the knife and spoon toward the center with clear space around it, and release it stable).
- **Coaching note:** keep it upright and put it on the center side of the setting.

Violation: Required alignment inspection skipped or incomplete

- **Visible cue:** the salt holder or centerpiece goes down before all four settings are inspected, or a visible placement or orientation error is left uncorrected.
- **SOP rule broken:** Step 6.1 (inspect Places 1–4 against every listed alignment and orientation check before placing the salt holder and centerpiece).
- **Coaching note:** run all eight inspection lines at all four places.

Violation: Entire setting dragged for alignment

- **Visible cue:** a loaded placemat is pushed or pulled to move the dishes, napkin, cutlery, and glass together.

- **SOP rule broken:** Step 6.2 (lift and replace only the failed item, then reinspect it; never align a complete setting by dragging its placemat).
- **Coaching note:** fix one failed object at a time.

Violation: Centerpiece staged incorrectly

- **Visible cue:** the centerpiece goes down before Step 6 passes or after the salt holder, is slid into position, is released unstable, or comes to rest touching a setting or covering the salt holder's spot at the front of the center zone. Contact made while carrying it in does not count, only where it ends up.
- **SOP rule broken:** Step 7 (place the centerpiece after alignment and before the salt holder: the left arm sets it in the handoff zone, the right gripper lowers it vertically at the center, and releases it upright and stable, resting clear of every setting).
- **Coaching note:** score the final resting position, not the trip in.

Violation: Salt holder placed incorrectly

- **Visible cue:** the salt holder goes down before the four settings pass Step 6 or before the centerpiece, lands tilted or unstable, sits on a setting instead of the center zone, or ends up touching the centerpiece.
- **SOP rule broken:** Step 8 (place the shared salt holder upright in the center zone after alignment and after the centerpiece, toward the front edge, clear of the centerpiece and every setting).
- **Coaching note:** salt holder last, upright, at the front of the center zone.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an item missing or misaligned, no salt holder, a failed centerpiece, something left in the handoff zone, an arm away from home, or no 3-second hold on the finished table.
- **SOP rule broken:** Step 9 (confirm the four aligned settings, salt holder, centerpiece, and empty handoff zone; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller-caused collision, or motor error.
- **Defective object:** a chipped, cracked, or bent item found mid-episode. Replace it before the next episode.

Annotation subtasks (from SOP)

1. Center one large plate on a placemat
2. Center one small plate on a large plate
3. Center one empty bowl on a small plate
4. Place one fork on its napkin
5. Place one knife beside a plate
6. Place one soup spoon outside a knife
7. Place one upright glass
8. Inspect and correct one setting
9. Place the centerpiece
10. Place the shared salt holder
11. Return both arms home and end the episode